2D brain plotting with ggseg :: Cheatsheet

Brain atlas data as a ggplot2 layer

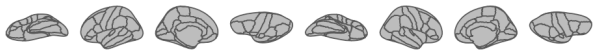
Plotting

One layer, one atlas

ggseg adds a single layer type to ggplot2. An atlas supplies the geometry, your data supplies the values.

```
library(ggseg)
library(ggplot2)

ggplot() +
  geom_brain(atlas = dk()) +
  theme_void()
```



- Atlases are **functions**: call `dk()`, not `dk`.
- Regions are grey until you map `fill` — see Atlas palettes.
- `theme_void()` removes the coordinate axes, which a brain plot has no use for.

```
geom_brain(
  mapping = aes(), data = NULL, atlas,
  hemi = NULL, view = NULL,
  position = position_brain(),
  context = TRUE, fun = mean, ...
)
```

Atlases that ship with ggseg

Atlas	Type	Covers
`dk()`	cortical	Desikan-Killiany, 35 regions per hemisphere
`aseg()`	subcortical	FreeSurfer automatic segmentation
`suit()`	cerebellar	SUIT cerebellar parcellation
`tracula()`	tract	TRACULA white-matter tracts

Dozens more install from the ggsegverse r-universe — see the atlases and data structures sheet.

Accessors live in **ggseg.formats**, which ggseg loads but does not re-export:

```
library(ggseg.formats)

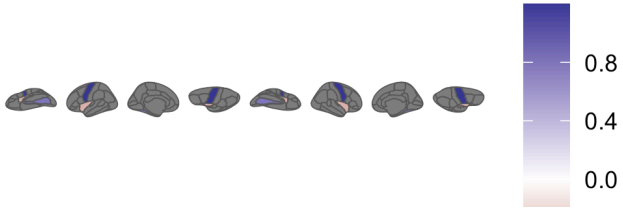
atlas_regions(dk())
atlas_labels(dk())
atlas_views(dk())
atlas_type(dk())
```

Your data on the brain

Give `ggplot()` a data frame with a `region` or `label` column. `geom_brain()` joins it to the atlas, so you only name the regions you actually have.

```
scores <- data.frame(
  region = c("precentral", "insula", "fusiform"),
  score = c(1.2, -0.4, 0.8)
)

ggplot(scores) +
  geom_brain(atlas = dk(), aes(fill = score)) +
  scale_fill_gradient2() +
  theme_void()
```



- Regions absent from your data are filled with `na.value` — grey by default.
- Join on `label` (`lh_precentral`) to colour one hemisphere only.
- `fun = mean` collapses several rows that share a region.
- `brain_join(data, atlas, by = NULL)` returns the merged table instead of a plot, when you want to inspect the join.

Facet by a grouping variable

Each facet gets a whole brain, so one plot can carry several contrasts.

```
long <- data.frame(
  region = rep(c("precentral", "insula", "fusiform"),
2),
  score = c(1.2, -0.4, 0.8, -1.1, 0.2, 1.6),
  session = rep(c("pre", "post"), each = 3)
)

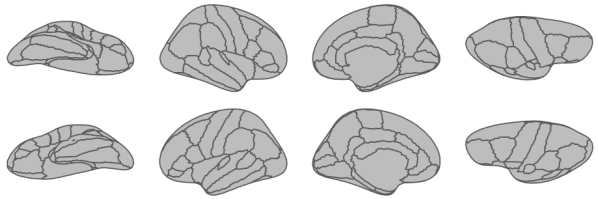
ggplot(long) +
  geom_brain(atlas = dk(), aes(fill = score)) +
  facet_wrap(~session) +
  theme_void()
```

Arranging

position_brain()

`position_brain()` decides where hemispheres and views sit. For cortical atlases, pass a **formula** of `rows ~ columns`.

```
ggplot() +
  geom_brain(
    atlas = dk(),
    position = position_brain(hemi ~ view)
  ) +
  theme_void()
```



Formula	Layout
`hemi ~ view`	hemispheres in rows, views in columns
`view ~ hemi`	views in rows, hemispheres in columns
`. ~ view + hemi`	a single row, every panel side by side
`hemi + view ~ .`	a single column, stacked

Zooming and trimming

```
# zoom each view onto its focus regions
position_brain(hemi ~ view, zoom = TRUE)

# zoom onto named regions, with 10% padding
position_brain(hemi ~ view, zoom = "insula", zoom_pad =
0.1)

# keep only these views, in this order
position_brain(views = c("lateral", "medial"))
```

Subcortical and tract atlases have slice views rather than hemispheres, so they take `"horizontal"`, `"vertical"`, `nrow` or `ncol` instead:

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```
ggplot() +
  geom_brain(
    atlas = aseg(),
    position = position_brain("horizontal", nrow = 2)
  ) +
  theme_void()
```



Picking hemispheres and views directly

```
geom_brain(atlas = dk(), hemi = "left")
geom_brain(atlas = dk(), view = "lateral")
```

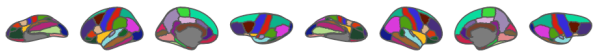
`reposition_brain(data, position, ...)` rearranges an already-joined atlas table, and `coord_brain(ratio = 1)` fixes the aspect ratio by hand.

Colour

Atlas palettes

Every atlas carries the colours its parcellation is normally drawn in. They are **not** applied automatically: map `fill` to `label` and hand the palette to `scale_fill_manual()`.

```
ggplot() +
  geom_brain(
    atlas = dk(),
    aes(fill = label),
    show.legend = FALSE
  ) +
  scale_fill_manual(values = dk()$palette) +
  theme_void()
```



`atlas_palette(dk())` returns the same named vector, and `atlas_plot_palette(dk())` draws it as a swatch.

Your own colours

```
pal <- c(
  precentral = "#5e3c58",
  insula = "#a8c5cb"
)

ggplot() +
  geom_brain(
    atlas = dk(),
    aes(fill = region),
    show.legend = FALSE
  ) +
  scale_fill_brain_manual(pal, na.value = "grey90") +
  theme_void()
```

- `scale_fill_brain_manual(palette, na.value)` — fill.
- `scale_colour_brain_manual()` / `scale_color_brain_manual()` — outlines.
- `scale_brain_manual(palette, aesthetics)` — both at once.

For continuous values, use ggplot2's own scales: `scale_fill_viridis_c()`, `scale_fill_gradient2()`, `scale_fill_distiller()`.

Deprecated. `scale_fill_brain()`, `scale_brain()` and the `*_brain2()` family are superseded. Use `scale_fill_manual()` with `atlas$palette`, or the `*_brain_manual()` functions above.

Themes

```
theme_brain(text.size = 12, text.family = "mono")
theme_darkbrain(text.size = 12)
theme_brain2(
  plot.background = "white",
  text.colour = "darkgrey"
)
theme_custombrain(
  plot.background = "white",
  text.colour = "darkgrey",
  text.size = 12
)
```

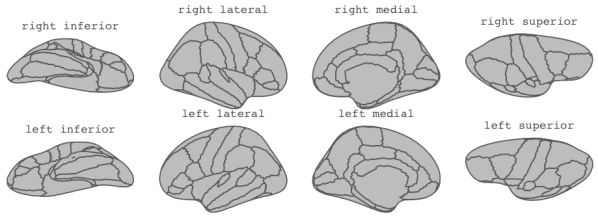
These keep the axes and style the surround; `theme_darkbrain()` is the one for dark slides. `ggplot2::theme_void()` is the right choice when you want nothing but the brain.

Labels and checks

Naming the views

```
pos <- position_brain(hemi ~ view)

ggplot() +
  geom_brain(atlas = dk(), position = pos) +
  annotate_brain(dk(), position = pos, size = 1.5) +
  theme_void()
```



- `annotate_brain()` writes the view name ("left lateral") above each view. Pass it the **same** `position` you gave `geom_brain()`.
- Subcortical and tract atlases get the raw view name instead ("axial_1", "sagittal").
- `scale_labs_brain()` and `scale_continous_brain()` put hemisphere and view labels on the axes in place of coordinates.
- Region names are not drawn by any of these — use `geom_sf_text()` on `as_sf_atlas(dk())` if you need them.

Checking an atlas

```
plot(dk())
brain_test_plot(
  dk(),
  position = position_brain(hemi ~ view)
)
```

Both draw every region in its palette colour — the quickest way to see whether a new or edited atlas renders.

Gotchas

- `dk` is a function; `dk()` is the atlas. The missing parentheses are the most common error.
- A brain with no `fill` mapping is grey by design, not broken.
- A region named in your data but missing from the atlas fails silently; check with `setdiff(data$region, atlas_regions(dk()))`.
- `context = FALSE` drops the greyed-out surrounding structures that subcortical atlases draw for anatomical context.
- `geom_brain_sf()` and `position_brain_sf()` are deprecated; `geom_brain()` is the polygon default, and `as_sf_atlas()` gives you `sf` geometry for `geom_sf()`.

Learn more

- `https://ggsegverse.github.io` — ecosystem documentation
- `vignette("ggseg")` — longer worked examples
- `https://github.com/ggsegverse` — source and issues

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